

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-16 22:38 UTC

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Generated 2026-05-16 22:38 UTC

- Cycles recorded: **716**
- Time span: 553.0h (~1.29 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	679
BARRIER_HIT	18
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	40
Schur-Weyl Duality	28
Free Probability Theory	19
Representation Theory of Symmetric Groups	18
Matroid Theory	16
Free Probability	14
Additive Combinatorics	13
Noncommutative L^p Geometry	10
Random Matrix Theory	8
Polymatroid Theory	8
Algebraic Geometry	7
Spectral Graph Theory	7
Finite Geometry	6
Plethysm Theory	6
Persistent Homology	5
Noncommutative Harmonic Analysis	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4

Field	Cycles
Algebraic Geometry over Finite Fields	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-13 07:16 UTC	INCONCLUSIVE	SOS Moment Matrix Eigenvalue Sum Bounded by Degree for Max-CUT
2026-05-13 07:23 UTC	INCONCLUSIVE	Symmetric Group Fourier Coefficient Count Inversely Proportional
2026-05-13 08:49 UTC	INCONCLUSIVE	Free Cumulant Sum Lower Bounded by Communication Complexity of DI
2026-05-13 09:19 UTC	INCONCLUSIVE	Schur-Weyl Component Count Gap in Monotone Circuit Size for Perma
2026-05-13 09:31 UTC	INCONCLUSIVE	Algebraic Shifting Generator Count Bounds Communication Complexit
2026-05-13 09:59 UTC	INCONCLUSIVE	Algebraic Shifting Edge Count Bounds Monotone DNF Size for k-CLIQ
2026-05-13 11:20 UTC	INCONCLUSIVE	Symmetric Square Irreducible Component Count Gap in Permanent vs
2026-05-13 11:47 UTC	INCONCLUSIVE	Symplectic Capacity Inverse Proportional to Extended Frege Proof
2026-05-13 12:08 UTC	INCONCLUSIVE	Symmetric Decomposition Coefficient Count Inversely Proportional
2026-05-16 20:13 UTC	INCONCLUSIVE	Layer-Commutator Frobenius Discrepancy Separates Read-Twice BP fo
2026-05-16 20:46 UTC	INCONCLUSIVE	Matroid Matching of Term Co-Occurrence Graph Bounds k-CLIQUE Mono

Time	Verdict	Title
2026-05-16 21:14 UTC	INCONCLUSIVE	Persistent H1 of Random Row Subclouds Bounds DISJ Communication
2026-05-16 21:44 UTC	INCONCLUSIVE	Non-Backtracking Spectral Gap Bounds Tseitin Resolution Length
2026-05-16 22:10 UTC	INCONCLUSIVE	Matching Polynomial Mahler Measure Lower-Bounds Tseitin DPLL Tree
2026-05-16 22:38 UTC	INCONCLUSIVE	Mobius-Mertens Cancellation on Fourier Levels vs Total Influence

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (sudo apt install pandoc texlive-xetex), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```